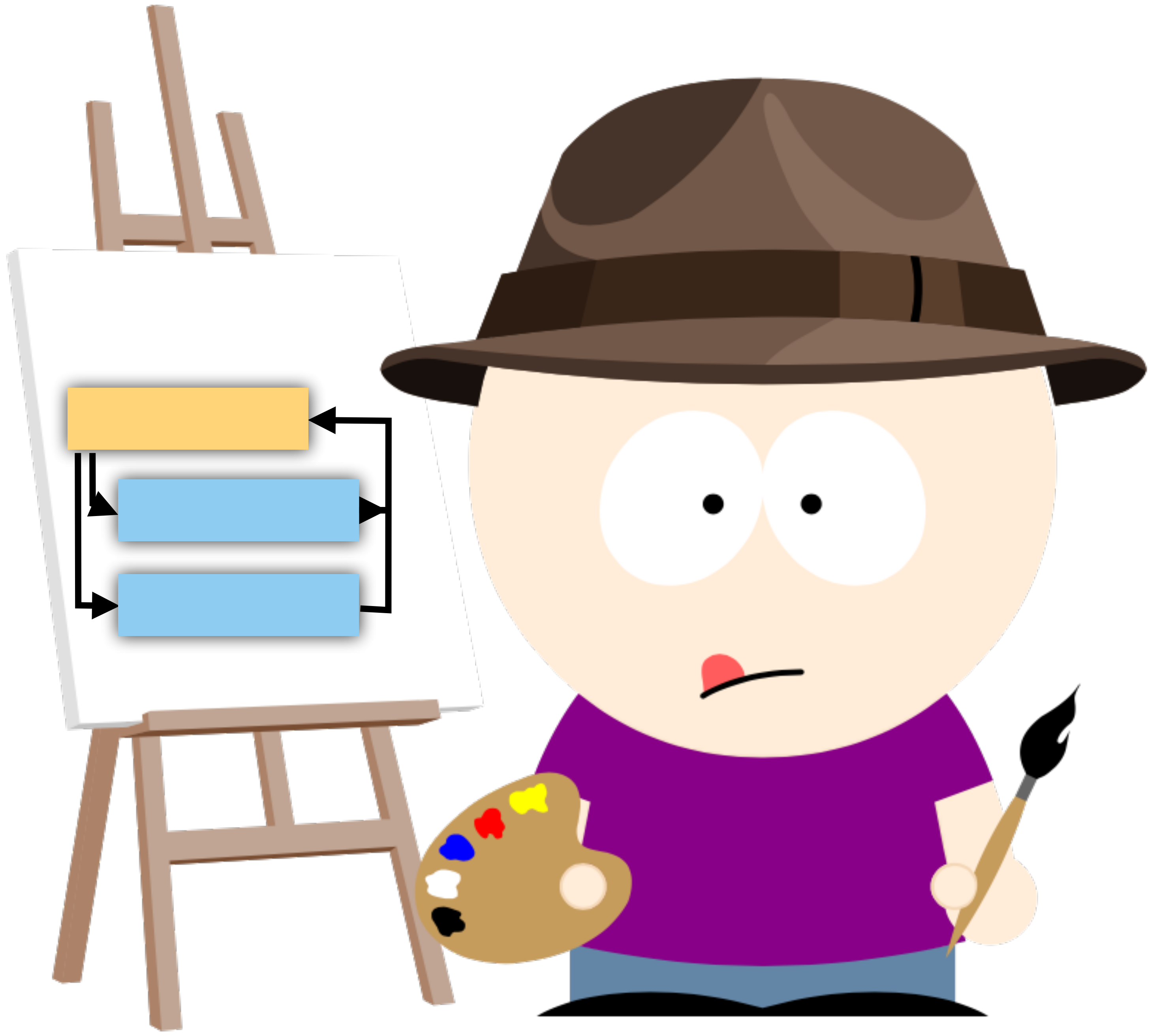

In-Class Exercises

Interpreters

Version: 2026-03-30

Download exercises

1. Go to

   https://ligerlabs.org/compilers.html

2. Download the file

   `interp-1-exercises.zip`

3. Open up a terminal and Unzip the file

```
> unzip interp-1-exercises.zip
> cd interp-1-exercises
> ls
```

Task 1

1. Consider the program in the file `task1.c`.

2. Without executing the program, show every step of the execution including the values of `pc`, `sp`, and `stack`.

3. Explain the macro `BINOP`!

```
#define BINOP(op,tp) {\
        stack[sp-2].tp = stack[sp-2].tp op stack[sp-1].tp; \
        sp--; \
        pc++; \
        break; \
    }
```

Task 2

1. Consider the program in the file `task2.c`.
2. Without executing the program, show every step of the execution including the evolution of the values of `i` and `result`.
3. What will the program print?
4. Explain what `&&` and `goto*` do.

Task 3

1. Consider the program in the file `task3.c`.

2. Without executing the program, show every step of the execution including the evolution of the values of `i` and `result`.

3. What will the program print?

4. Explain the declaration

```
void(*calls[])() = {foo,foo,foo,baz,foo,foo,foo,bar,zap,0x0};
```

Task 4

1. Consider the program in the file `task4.c`.

2. Explain the `foo` declaration.

3. Explain what each of the functions `bar`, `baz`, `zap`, and `quux` do.

4. Without executing the program, what will it print?

```
struct foo {
    unsigned char tag;
    union {
        int i;
        float f;
    };
};
```

Task 5

1. Consider the program in the file `task5.class`. It the result of compiling a program `task5.java`, which we have unfortunately lost!
2. However, I was able to use the javap program to recover the Java byte codes for the function `foo`. You can see them to the right.
3. What does `foo` do? Show the original Java source for `foo`!

```
 0: bipush 6
 2: istore_1
 3: bipush 7
 5: istore_2
 6: iload_1
 7: iconst_2
 8: iadd
 9: iload_2
10: iconst_5
11: iadd
12: imul
13: istore_3
14: iload_3
15: ireturn
```

- Learn about Java byte code instructions here:

https://en.wikipedia.org/wiki/List_of_JVM_bytecode_instructions

Task 6

1. Consider the program in the file `task6.c.`

2. Without executing the program, show every step of the execution including the evolution of the values of `i` and `result`.

3. What will the program print?

4. Explain what this statement does:

```
goto* jumps[jTab[i]];
```

Task 7

1. Consider the program in the file `task7.class`.
2. To the right are the Java byte codes for the function `foo`. What does `foo` do? Show what you think is the original Java source for `foo`!

```
 0: bipush     6
 2: istore_1
 3: iload_1
 4: bipush   100
 6: if_icmpge 18
 9: iload_1
10: iconst_2
11: imul
12: iconst_1
13: iadd
14: istore_1
15: goto       3
18: iload_1
19: ireturn
```

- Learn about Java byte code instructions here:

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Definitions and Algorithms

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